

MASTER'S COMPREHENSIVE EXAMINATION

Part II Applied (2 hours) April 1st, 2017.

Instructions: Attempt any three out of the following four questions on both pages. Mention which three you would like to have graded. Open book and open notes.

1) A researcher wants to see the effect of different types of music in alleviating depression in patients. The dependent variable is the Improvement in Mood measured on a scale that is normally distributed for the population studied. In an experiment, 6 male subjects and 6 female subjects are picked at random from the population. 2 males, and 2 females are randomly assigned to listen to rock music; 2 males, and 2 females are randomly assigned to listen to classical music; the remaining 2 males, and 2 females are assigned to listen to Jazz music. Each subject is measured on the Improvement in Mood after a morning (9 to 11 am) listening session, after an afternoon (3 to 4pm) listening session, and after an evening (7 to 9 pm) session. A subject listens to the same type of music for all three sessions,

a) Show the data layout for the experiment.

b) Show the ANOVA table filling in the columns for Source of Variability, degree of freedom, Expected Mean Square, and denominator of F-ratio (for testing Source of Variability).

2. THE TOWERS OF THANG-LONG are ceramics that were produced by only two schools in the 14th century, the Borneo-cast school and the Malacca-cast school. It is very difficult to determine which school had manufactured a given ceramic tower but it is known that:



- Height of towers made by the Borneo-cast are distributed $N(25.00, 10.00)$
- Height of towers made by the Malacca-cast are distributed $N(30.00, 5.00)$.
- Also 40% of all Thang-Long towers ever made were made by the Borneo cast.

- An archeologist discovers a Thang-Long tower that is 27.50 units. Use the above information to estimate the probability that it was made by the Borneo-cast school
 - An archeologist has 5 Thang-Long towers that were all made by the same school their mean height is 27.50 units. Use this to estimate the probability that they were made by the Borneo-cast school.
 - What additional information from the measurements of these 5 Thang-Long towers in b could help to produce a better estimate for the probability that it was made by the Borneo-cast school. Only describe the additional information needed; no calculations are expected
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3. A Spanish to English dictionary has 200 pages, each page containing 40 entries. Each entry gives a Spanish word followed by its English translation. A Spanish teacher wants to carry out a hypothesis test to see if a student knows the translation of more than half the words in that dictionary. The teacher picks a simple random sample with replacement of 6 pages from the 200 pages. The teacher asks the student to translate all 40 words on each chosen page. (a total of 240 words). The teacher records for each sampled page the number of Spanish words on the page for which the student knew the correct English translation. The following are the number of words the student knew correctly for 6 chosen pages: 13, 36, 37, 13, 12, 15.

a) Test H_0 : that the student does not know the correct translation of more than half the words in the (200 pages of the) dictionary), against the alternative at the 0.05 level of significance. Show the formula for the test statistic used, and the substitution of numbers in the formula.

b) Estimate the **total number** of Spanish words that the student knows the correct English translation in the (200 pages of the) dictionary. Give the standard error of this estimate. Show formula used and substitution in formula.

4. THE SNAKES OF THE KALAHARI undergo a long migration during the summer months where many of the snakes die. Dr. Taylor measured pre-migration weight centigrams (cg) and length in centimeters (Cm) on 100 snakes plus whether or not those snakes survived the migration.



SHE FIRST FITS THIS **LOGISTIC REGRESSION**
MODEL FOR SURVIVING MIGRATION THAT USES WEIGHT AS A PREDICTOR

	Estimate	Variance	
Intercept ($\hat{a}$)	-0.7	0.25	$Cov(\hat{a}, \hat{b}) = -0.007$
Weight ($\hat{b}$)	0.2 /cg	0.0004	

A. If Snake A is 2 cg heavier than Snake B. Give an odds ratio and 95% confidence interval for the odds ratio of Snake A surviving the migration Vs. Snake B.

B. Give a point estimate and 95% confidence interval for the probability that a 10 cg snake survives the migration

SHE THEN FITS TWO MORE MODELS FOR SURVIVING THE MIGRATION WITH [LENGTH] AND [LENGTH + WEIGHT] AS PREDICTORS

MODEL WITH ONLY LENGTH			MODEL WITH LENGTH + WEIGHT		
	Estimate	Variance		Estimate	Variance
Intercept	1.3	0.32	Intercept	-0.4	0.50
Length	-0.05	0.004	Length	-0.6	0.003
			Weight	0.3	0.0003

C. Use the information on all three models fitted to determine the best answer for if there is a causal association between length and probability to survive